

Ex-Ante ERW — Results dashboard

Profitability maps and tables under different assumptions

2026-09-01

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Profitability under different assumptions

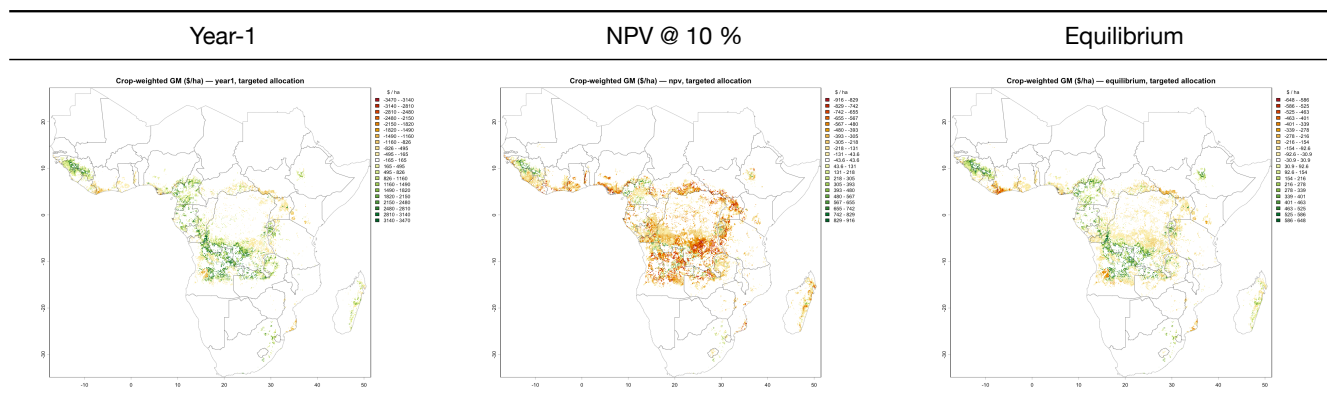
This is a focused results dashboard. It complements `erw-model.md` (full methodology) and `erw-tutorial.md` (worked example). All figures and tables below are generated from the 276 gross-margin rasters under `data/economics_erw/` by `erw/erw-10-profitability-maps-tables.R`.

Scope: Sub-Saharan Africa, 23 SPAM crops, 0.083° working grid. **Regimes:** year-1 (one application, agronomic + CDR revenue in year 1) · NPV @ 10 % (CDR revenue phased over 5 years, agronomic returns over the reapplication interval) · equilibrium (steady-state maintenance dose). **Allocations:** targeted (LiTAS dose only on acid pixels above each crop's tolerance) · `uniform_10` / `uniform_20` / `uniform_50` (uniform t basalt / ha across all cropland regardless of soil acidity). **Model basis (refreshed 2026-09-01):** Arrhenius temperature factor ($E_a = 68.8$ kJ/mol, headline; linear retained as sensitivity), triangular soil-pH factor peaking at **pH 6.0** (was 5.0), yield-sensitivity sweep capped at 1.8×. All maps and tables below are from the 2026-09-01 run. **Default parameters (set 2026-05-26):** carbon price \$150 / tCO₂, MRV **\$20 / tCO₂** (lowered from \$30), **grain size 50 μm** (lowered from 100 μm; the Streffer grinding-curve anchor), \$10 / t quarry gate, country-level electricity tariffs and grid CI, spatial delivered-transport costs, year-1 agronomic return **summed over the 5-year basalt residency** (was a single-season figure), CGIAR-CSI Global Aridity Index for the pedogenic deduction, per-crop EcoCrop max-pH cliffs (range 5.5–8.0).

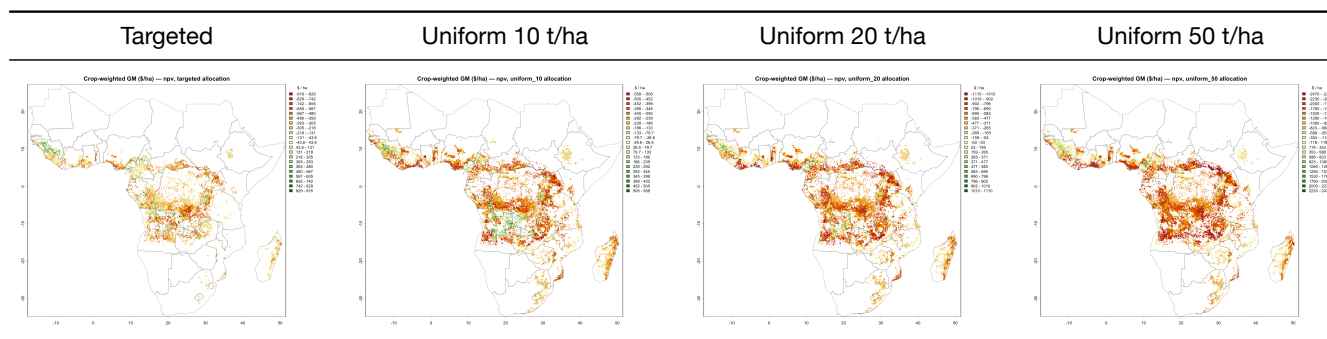
1. Continent-wide gross margin

Each map below shows the crop-weighted average gross margin per hectare (\$ / ha) at the 0.083° pixel — weighted by harvested area across the 23 crops sharing that pixel. Green pixels are profitable; red are loss-making.

1.1 By return regime (LiTAS-targeted allocation)



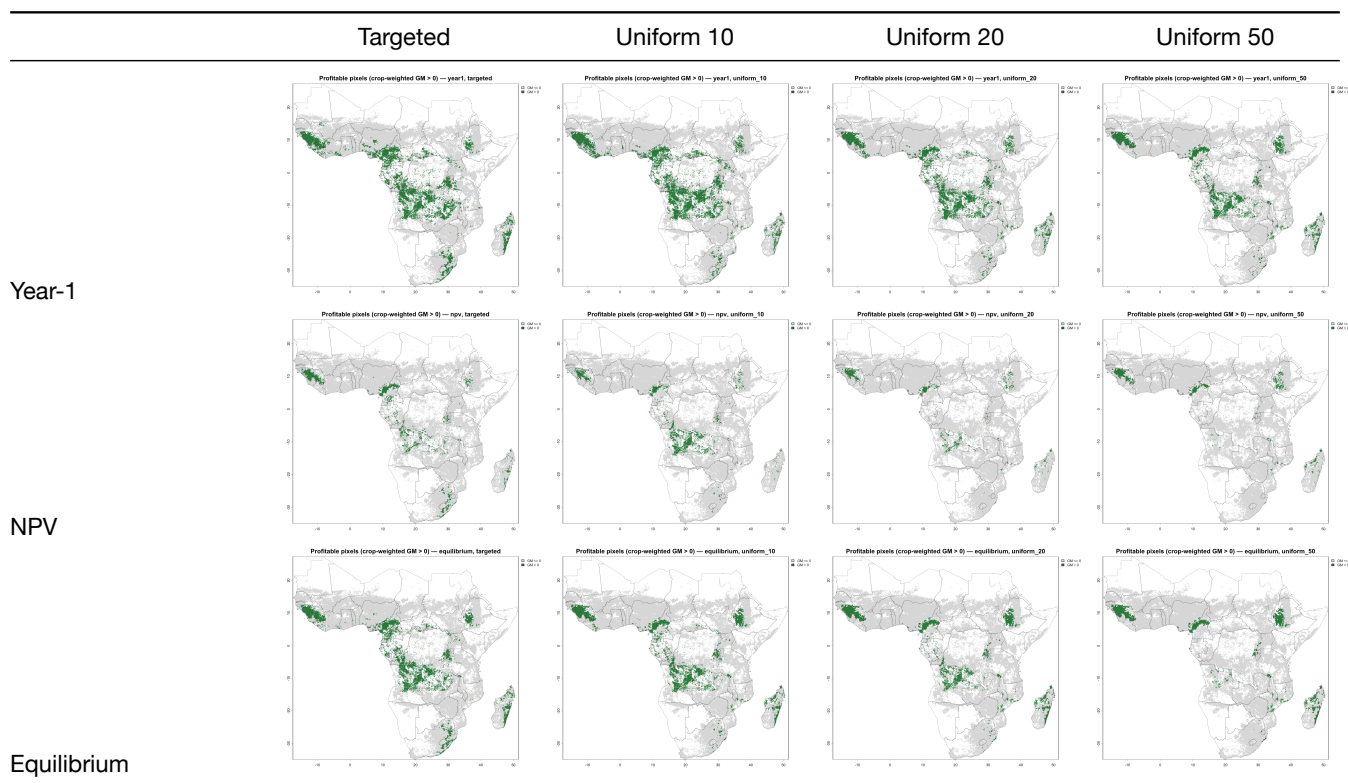
1.2 By allocation rule (NPV regime)



The targeted allocation prunes the action set to acidic pixels and pays the LiTAS-calibrated dose; uniform rules pay a fixed rate across all cropland and recover CDR even on near-neutral soils that the targeted rule skips.

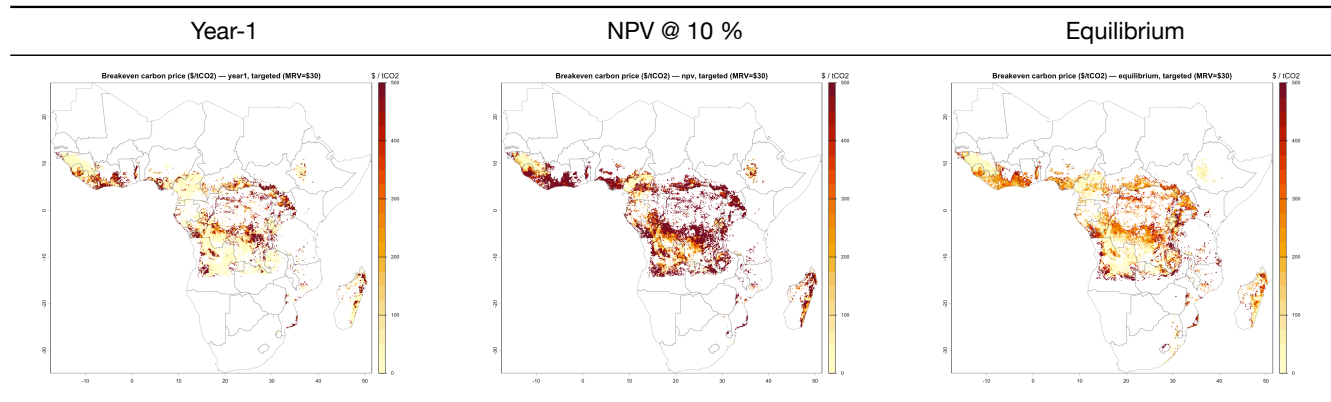
2. Profitable footprint

Binary masks showing where the crop-weighted GM is strictly positive. These are the candidate pixels for the supply-ranked deployment in erw-supply.



3. Breakeven carbon price

For each pixel, the minimum carbon price (\$ / tCO₂) that drives crop-weighted GM to zero, holding all other defaults constant (MRV \$20 / tCO₂, grain size 50 µm, etc.). Pixels with no CDR potential are masked; the legend caps at \$500 / tCO₂ for legibility.



The equilibrium regime tends to need higher carbon prices because the maintenance dose recurs forever, while year-1 / NPV amortise the up-front agronomic uplift against a single application.

4. Per-crop gross margin (selected crops, targeted)

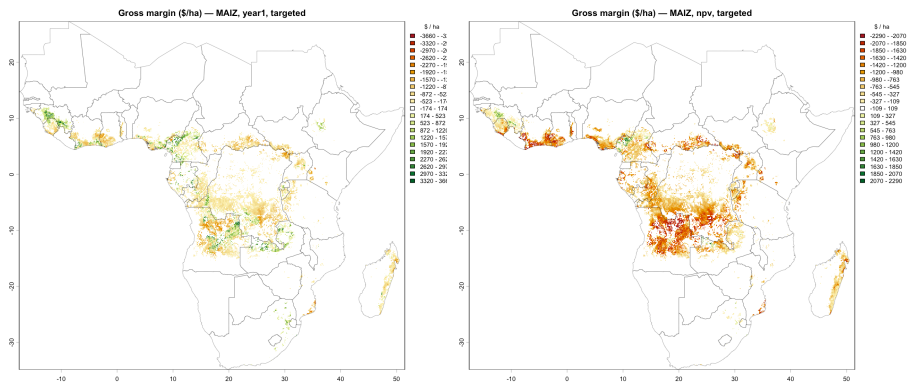
The five highest-acreage crops, year-1 and NPV regimes:

Crop

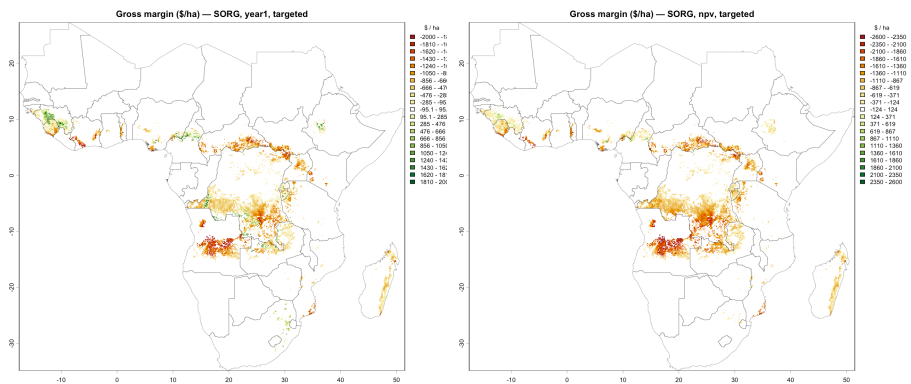
Year-1

NPV @ 10 %

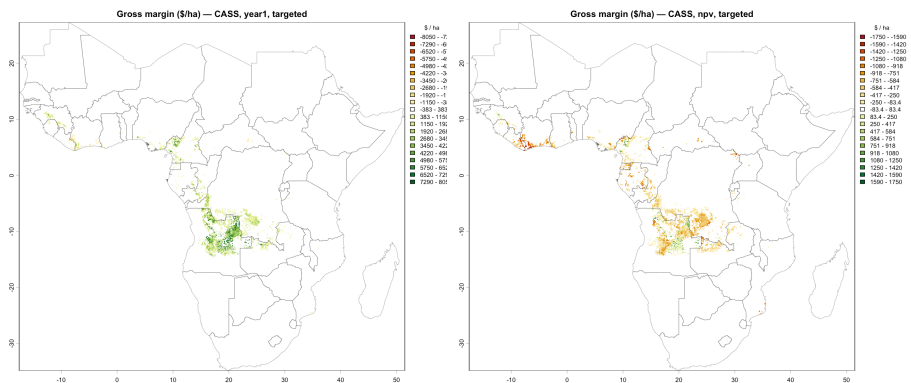
Maize (MAIZ)



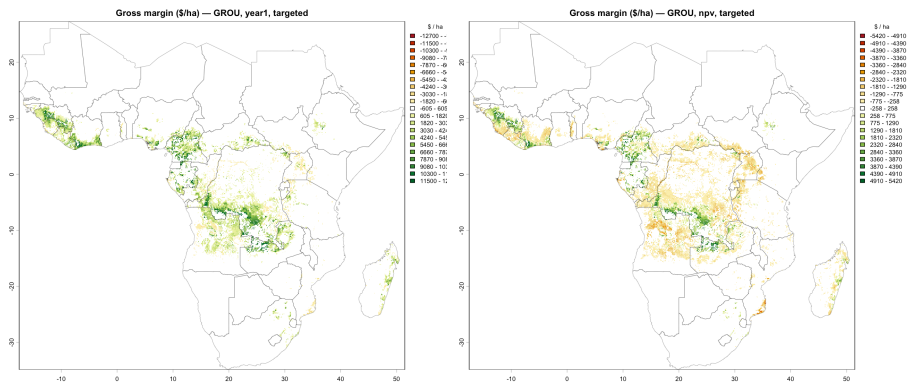
Sorghum (SORG)



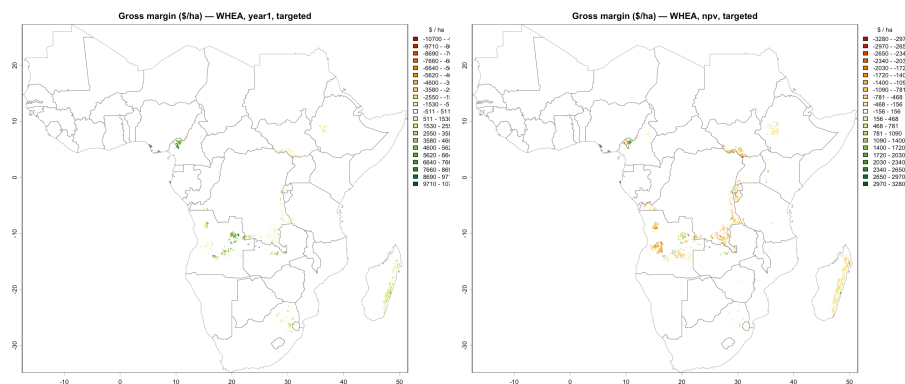
Cassava (CASS)



Groundnut (GROU)



Wheat (WHEA)



5. SSA-wide totals — one row per regime × allocation

Aggregates are summed across all 23 crops, all pixels (no positive-GM filter unless the column says “profitable”). Source: docs/tables/output-ssa-summary.csv.

Regime	Allocation	Total GM (M\$)	Agro return (M\$)	Basalt cost (M\$)	CDR revenue (M\$)	Total CDR (Mt)	Profitable area (Mha)	CDR on profitable (Mt)
year1	targeted	12,371	30,220	21,545	4,483	34.5	7.36	25.6
year1	uniform_10	14,357	30,220	75,918	11,512	88.6	10.61	1.1
year1	uniform_20	1,422	30,220	151,836	27,978	215.2	8.07	7.6
year1	uniform_50	-25,337	30,220	379,591	92,169	709.0	6.65	50.5
npv	targeted	-5,926	12,030	21,545	3,560	34.5	2.55	16.9
npv	uniform_10	-10,548	5,494	75,918	9,142	88.6	3.23	0.4
npv	uniform_20	-24,285	5,494	151,836	22,218	215.2	1.86	2.4
npv	uniform_50	-55,929	5,494	379,591	73,196	709.0	1.98	19.3
equilibrium	targeted	1,772	6,044	7,364	3,248	25.0	6.45	12.4
equilibrium	uniform_10	-3,774	6,044	75,918	17,967	138.2	7.75	15.7
equilibrium	uniform_20	-11,850	6,044	151,836	40,188	309.1	6.32	29.2
equilibrium	uniform_50	-36,049	6,044	379,591	107,153	824.2	4.92	63.2

6. Top crops by aggregate gross margin (NPV regime)

The full crop × regime × allocation matrix is in docs/tables/output-crop-ranking.csv. Below: the top eight crops by total GM under the NPV-targeted scenario.

Rank	Crop	Total GM (M\$)	Profitable area (Mha)	Total CDR (Mt)
1	POTA	731	0.26	1.42
2	BEAN	594	0.78	10.41
3	GROU	435	0.68	7.57
4	LENT	18	0.01	0.04
5	CHIC	-0	0.01	0.06
6	TEAS	-1	0.00	0.00
7	TOBA	-6	0.01	0.02

Rank	Crop	Total GM (M\$)	Profitable area (Mha)	Total CDR (Mt)
8	BARL	-13	0.00	0.07

7. Country breakdown — NPV, targeted (top 12 by GM)

Full per-country table for all regime × allocation combinations: docs/tables/output-country-summary.csv (528 rows = 44 countries × 12 scenarios). Below: the top dozen by aggregate GM in the NPV-targeted scenario.

Rank	ISO3	GM (M\$)	Profitable area (Mha)	Total CDR (Mt)
1	CMR	2,602	0.64	10.66
2	GIN	362	0.32	1.99
3	RWA	103	0.23	0.50
4	ZAF	19	0.06	0.00
5	BWA	0	0.00	0.00
6	DJI	0	0.00	0.00
7	ERI	0	0.00	0.00
8	GMB	0	0.00	0.00
9	MRT	0	0.00	0.00
10	NAM	0	0.00	0.00
11	SDN	0	0.00	0.00
12	SOM	0	0.00	0.00

8. How to rebuild

1. Make sure erw-7 outputs exist

```
ls data/economics_erw/targeted/ # should have 69 = 23 crops × 3 regimes
```

2. Render maps + tables

```
Rscript erw/erw-10-profitability-maps-tables.R
```

3. Rebuild this PDF

```
bash docs/build-erw-results.sh
```

The R script is idempotent and ~3 min on an M-series laptop. The PDF build requires pandoc + xelatex (same toolchain as erw-model.pdf).